Data	STEM (A) Sho	Non Stem (B)	di
1	7	7	0
2	9	10	1
3	5	6	1
4	7	4	3
5	7	7	0
6	7	12	5
7	4	5	1
8	7	8	1
9	5	4	1
10	13	6	7
11	12	7	5
12	5	7	2
13	5	7	2
14	4	11	7
15		9	9
16		10	10
17		7	7
18		7	7

Hypothesis:Does the data in the table indicate a difference in showers showers for STEM and Non-STEM Majors?

d_	69	/18	3.8333333
Sd	3.2943802		
SE= (Sd/(n)^.5)	0.7764929		
Test stat	4.936727		

M0=0

Ma= Avg From A

Mb= Avg from B

H0: Ma-Mb=0

Ha: Ma-Mb≠0

Df= n-1 17

Two tail test

alpha =5%

alpha/2= 2.5%

1-alpha/2 97.500%

Crit Value approach

Crit Value: 2.1098156

Crit Value		T Stat
2.1	<	4.9

We reject H0

P-Value Approach

1-tail prob	0.9999374
tail prob=	6E-05
p_value	1E-04 which is about 0.0001

Pvalue		Alpha
0.10%	<	5%

We Reject Ho

What does this mean?

Based on the data, we can indicate a differnce in the mean for STEM and NON STEM majors. Since the P-Value was significantly smaller than 5%, we can assume there is a significant differnce in the showers between STEM and non STEM majors

t-Test: Two-Sample Assuming Unequal Variances

	STEM (A) Showe	*Stem (B) Showers*
Mean	6.9285714	7.4444444
Variance	7.6098901	4.9673203
Observations	14	18
Hypothesized M	0	
df	25	
t Stat	-0.5698512	
P(T<=t) one-tail	0.2869322	
t Critical one-tail	1.7081408	
P(T<=t) two-tail	0.5738643	
t Critical two-tail	2.0595386	

We fail to reject H0

Meanining there is not likeley a differnce between STEM major and Non Stem Major shower patters as

P value> alpha (5%)

and Crit values are greater than T stat